Scam & Spam Message Detection Using Machine Learning and AI

1. Introduction

Scam and spam messages are a common problem in digital communication. These messages may contain misleading offers, suspicious links, fake prizes, or requests for personal information.

The **Scam & Spam Message Detection** project is designed to analyze text messages and classify them as spam or ham (ordinary messages) using Machine Learning. A Streamlit web application provides a simple interface where users can enter a message and view the model's prediction. An optional Gemini AI integration provides additional explanations when the service is available.

2. Problem Statement

Manually identifying suspicious messages can be difficult and time-consuming. Some spam messages appear genuine and may trick users into clicking unsafe links or sharing sensitive information.

This project aims to develop a message classification system that identifies potential spam messages and highlights suspicious warning signs.

3. Project Objectives

The main objectives are:

- To classify messages as spam or ham.
- To preprocess text messages before classification.
- To apply TF-IDF for text feature extraction.
- To use Logistic Regression for message classification.
- To evaluate the model using accuracy, precision, recall, and F1-score.
- To build a user-friendly application using Streamlit.
- To provide optional AI-generated explanations through Gemini.

4. Dataset Description

The project uses the SMS Spam Collection dataset.

- **Total messages:** 5,574
- **Ham messages:** 4,827
- **Spam messages:** 747
- **Dataset columns:** `message` and `label`

The dataset contains ordinary messages labelled as `ham` and unwanted messages labelled as `spam`.

5. Tools and Technologies

- **Python:** Main programming language.
- **Pandas:** Dataset loading and handling.
- **Scikit-learn:** Machine Learning model and evaluation.
- **TF-IDF:** Converts text into numerical features.
- **Logistic Regression:** Classifies messages as spam or ham.
- **Streamlit:** Creates the web application interface.
- **Matplotlib:** Generates the confusion matrix graph.
- **Google Gemini API:** Provides optional explanations of messages.
- **Visual Studio Code:** Development environment.

6. Methodology

The project follows these steps:

1. **Data Loading:** Load the SMS dataset from a CSV file.
2. **Text Preprocessing:** Convert text to lowercase, replace URLs, remove unwanted characters, and normalize whitespace.
3. **Feature Extraction:** TF-IDF represents text messages as numerical features.
4. **Model Prediction:** The saved TF-IDF and Logistic Regression pipeline predicts whether a message is spam or ham.
5. **Warning Sign Detection:** The application checks for words such as “urgent,” “free,” “prize,” “verify,” and “OTP.”
6. **AI Explanation:** Gemini may provide additional context and safety recommendations when available.
7. **Evaluation:** The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

7. Model Evaluation

The evaluation script loaded 5,574 messages and evaluated the saved model on a stratified 20% split containing 1,115 messages.

Overall Result

- **Accuracy:** 99.19%
- **Evaluation messages:** 1,115

Classification Report

Metric	Ham	Spam
Precision	1.00	0.95
Recall	0.99	0.99
F1-score	1.00	0.97
Support	966	149

The displayed classification report rounds the metrics to two decimal places.

Interpretation

The model achieved 99.19% accuracy on the evaluated split. Spam precision was 95%, indicating that some messages predicted as spam may be ordinary messages. Spam recall was 99%, indicating that the model identified most spam messages in this evaluation.

Evaluation note: The saved model's original training split has not yet been verified. Therefore, these results should be treated as preliminary until it is confirmed that the evaluation messages were not used to train the saved model.

8. Confusion Matrix

A confusion matrix compares the model's predictions with the actual labels.

The project generates a graph named `confusion_matrix.png`. It helps visualize correct and incorrect classifications for both ham and spam messages.

Insert the `confusion_matrix.png` graph here.

9. Application Features

The Streamlit application includes:

- A text area for entering messages.
- A button to analyze messages.
- A spam or ham prediction.
- Detection of selected warning words.
- Optional Gemini AI explanations.
- Error handling when the AI service is unavailable.
- A safety disclaimer advising users not to share passwords, banking details, or OTPs with untrusted sources.

10. Limitations

- The model may incorrectly classify some messages.
- New scam techniques and unfamiliar wording may reduce performance.
- The warning-word checker only recognizes a limited set of patterns.
- Gemini explanations depend on API availability.
- The reported evaluation score needs confirmation against the model's original training split.
- A message classified as ham is not guaranteed to be safe.

11. Future Improvements

Possible improvements include:

- Evaluating the model on a verified, independent test dataset.
- Expanding the dataset with more recent scam messages.
- Improving text preprocessing and warning-sign detection.
- Adding a message history and downloadable results.
- Comparing Logistic Regression with other classification algorithms.
- Improving the application's usability and accessibility.

12. Conclusion

The Scam & Spam Message Detection project demonstrates how Machine Learning and AI can help analyze suspicious text messages. The TF-IDF and Logistic Regression pipeline classifies messages, while the Streamlit interface makes the system accessible to users. Optional Gemini integration adds explanations when the service is available.

The evaluation script reported an accuracy of 99.19% on its test split. Further validation is necessary to confirm that this score reflects performance on genuinely unseen messages. The application should be used as a warning and assistance tool rather than a guarantee that a message is safe.

13. References

1. UCI Machine Learning Repository — SMS Spam Collection Dataset: https://archive.ics.uci.edu/dataset/228/sms+spam+collection
2. Scikit-learn Documentation: https://scikit-learn.org/
3. Streamlit Documentation: https://docs.streamlit.io/
4. Google Gemini API Documentation: https://ai.google.dev/gemini-api/docs